1. Users

The purpose of this collection is to store information about the user, including their login credentials and account details. Users can fall under one of three types: ‘admin’, ‘tenant’, and ‘landlord’.

```
{
  "_id": "user-aa11bb22-cc33-dd44-ee55-ff6677889900",
  "firstName": "Jane",
  "lastName": "Doe",
  "ownedProperties": [
    "prop-aa11bb22-cc33-dd44-ee55-ff6677889900",
    "prop-bb22ff55-xx54-fd67-hn54-jh4466883311"
  ],
  "savedProperties": [
    "prop-bb22ff55-xx54-fd67-hn54-jh4466883311",
    "prop-aa11bb22-cc33-dd44-ee55-ff6677889900"
  ],
  "email": "JDoe123@gmail.com",
  "hashedPassword": "$123abc$456$def",
  "userRole": "tenant",
  "phoneNumber": "+2675558888"
}
```

Name	Type	Description
_id	string	A globally unique identifier to represent the user.
firstName	string	The user’s first name.
lastName	string	The user’s last name.

ownedProperties	array	An array containing the corresponding IDs for each of the user's owned properties.
savedProperties	array	An array containing the corresponding IDs for each of the user's saved properties.
email	string	The user's email, for login credentials.
hashedPassword	string	The user's password, encrypted, for login credentials.
userRole	string	A string constant that indicates the role of the account. This can be 'tenant', 'landlord', or 'admin'.
phoneNumber	string	A string containing the user's phone number (for use with notifications).

2. Properties

The purpose of this collection is to store information about properties, including address information, timestamping, and any reviews or violations associated with them.

```
{
  "_id": "prop-aa11bb22-cc33-dd44-ee55-ff6677889900",
  "address": {
    "number": "123",
    "street": "Maple Road",
    "city": "New York City",
    "state": "New York",
    "zipCode": "10001"
  },
  "violations": [
```

```
    "viol-00112233-4455-6677-8899-aabbccddeeff",
    "viol-11667788-7711-4488-0011-wwccqqjjkkll"
  ],
  "createdOn": "2026-01-01-0000",
  "updatedOn": "2026-01-02-0000",
  "reviews": [
    "67e5b1d9a8c41234abcd3001",
    "45f9h4lfdsfa0vnbk4993aasdf"
  ]
}
```

Name	Type	Description
_id	Object	A globally unique identifier to represent the property.
address	Subdocument	A subdocument which stores the address of the property.
violations	array	An array storing the corresponding violation IDs for all active violations on the property.
createdOn	date	The date that the listing for this property was created.
updatedOn	date	The date that the listing for this property was last updated.
reviews	array	An array containing all of the corresponding review IDs for this property.

Address (Subdocument)

Address is a subdocument that stores the individual components of an address (for a property)

Name	Type	Description
number	number	Street number of the property
street	string	Street name of the property
city	string	City of the property
state	string	State of the property
zipCode	string	Zip code of the property.

3. Violations

The **Violations** collection stores all housing violations associated with properties in the application. It contains data imported from the NYC housing dataset as well as application-specific fields used for tracking deadlines, countdown alerts, remediation workflows, and filtering.

This collection directly supports core application features such as the violation dashboard, expiration countdowns, violation table pages, and landlord/tenant remediation updates. It enables users to monitor active violations, track deadlines, and view the current status of each issue.

```
{
  "_id": "67e5b1d9a8c41234abcd1001",
  "datasetViolationId": "HPD-2026-004219",
  "propertyId": "67e5b1d9a8c41234abcd2001",
```

```
  "buildingAddress": "123 Main St",
  "normalizedAddress": "123 main st manhattan ny 10001",
  "borough": "Manhattan",
  "zipCode": "10001",
  "violationType": "Mold",
  "violationDescription": "Black mold detected in bathroom ceiling
ventilation area",
  "violationClass": "B",
  "violationStatus": "Open",
  "originalCertifyByDate": "2026-04-15T00:00:00.000Z",
  "inspectionDate": "2026-03-20T00:00:00.000Z",
  "repairScheduledAt": "2026-03-27T15:00:00.000Z",
  "resolvedAt": null,
  "daysRemaining": 19,
  "isActionable": false,
  "source": "NYC Open Data",
  "lastDatasetSyncAt": "2026-03-27T14:30:00.000Z",
  "lastUpdatedByUserId": "67e5b1d9a8c41234abcd3001",
  "remediationStatus": {
    "currentState": "Repair Scheduled",
    "updatedByUserId": "67e5b1d9a8c41234abcd3001",
    "updatedAt": "2026-03-27T15:00:00.000Z",
    "notes": "Landlord scheduled contractor visit for April 1"
  },
  "statusHistory": [
    {
      "state": "Open",
      "changedAt": "2026-03-20T00:00:00.000Z",
      "changedBy": "system"
    },
    {
      "state": "Repair Scheduled",
      "changedAt": "2026-03-27T15:00:00.000Z",
      "changedBy": "67e5b1d9a8c41234abcd3001"
    }
  ],
  "createdAt": "2026-03-20T00:00:00.000Z",
  "updatedAt": "2026-03-27T15:00:00.000Z"
}
```


Field Definitions

Name	Type	Description
_id	string	A globally unique identifier for the violation.
datasetViolationId	string	The original violation identifier from the NYC dataset.
propertyId	string	The id of the property associated with the violation.
buildingAddress	string	The street address of the property.
normalizedAddress	string	A standardized version of the address used for efficient searching and lookup.
borough	string	The NYC borough where the property is located.
zipCode	string	The ZIP code of the property.
violationType	string	Category of violation such as mold, heat, or water.
violationDescription	string	Detailed description of the violation.
violationClass	string	Official class of violation (A, B, or C).
violationStatus	string	Current violation status such as Open, Closed, Repair Scheduled, or Resolved.
originalCertifyByDate	date	Deadline by which the violation must be corrected.
inspectionDate	date	Date the violation was inspected.
repairScheduledAt	date	Date when a repair was scheduled for the violation.
resolvedAt	date	Date when the violation was resolved. Used for calculating resolution time.
daysRemaining	number	Number of days remaining before the deadline expires.
isActionable	boolean	Indicates whether the violation is past its deadline and actionable.
source	string	Source of the violation data, such as NYC Open Data.
lastDatasetSyncAt	date	Last time the violation was synced from the dataset.
lastUpdatedByUserId	string	The id of the user who last updated the violation.
remediationStatus	subdocument	Stores the current remediation state and metadata.
statusHistory	array of subdocuments	Stores a timeline of status changes.

createdAt	date	Date the record was created in the application.
updatedAt	date	Date the record was last updated.

RemediationStatus

The **RemediationStatus** subdocument stores the most recent remediation or dispute state for a violation. This allows the application to display real-time updates for landlord or tenant actions.

```
{
  "currentState": "Repair Scheduled",
  "updatedByUserId": "67e5b1d9a8c41234abcd3001",
  "updatedAt": "2026-03-27T15:00:00.000Z",
  "notes": "Landlord scheduled contractor visit for April 1"
}
```

Name	Type	Description
currentState	string	Current remediation state such as Repair Scheduled, Resolved, or Disputed.
updatedByUserId	string	User who last updated the remediation status.
updatedAt	date	Timestamp of the latest remediation update.
notes	string	Additional notes about the remediation progress or dispute.

StatusHistory

The StatusHistory subdocument tracks all changes made to a violation status over time. It provides an audit trail so the application can show how a violation progressed from open to repair scheduled, resolved, or disputed.

```
{
  "state": "Repair Scheduled",
  "changedAt": "2026-03-27T15:00:00.000Z",
  "changedBy": "67e5b1d9a8c41234abcd3001"
}
```

Field Definitions

Name	Type	Description
state	string	Status of the violation at a specific point in time.
changedAt	date	Timestamp of the status change.
changedBy	string	User or system responsible for the status change.

4. Evidence

The **Evidence** collection stores user-submitted documentation associated with violations. This includes uploaded photos, notes, timestamps, and metadata linking each record to a violation and property. This collection supports the Evidence Vault feature and allows users to build a verifiable, time-stamped record that can later be exported for use in Housing Court.

```
{
  "_id": "67e5b1d9a8c41234abcd4001",
  "violationId": "67e5b1d9a8c41234abcd1001",
  "propertyId": "67e5b1d9a8c41234abcd2001",
  "uploadedByUserId": "67e5b1d9a8c41234abcd3002",
  "evidenceType": "photo",
  "fileName": "bathroom_mold_03272026.jpg",
  "fileUrl": "/uploads/evidence/bathroom_mold_03272026.jpg",
  "mimeType": "image/jpeg",
  "fileSize": 2451821,
  "caption": "Mold still visible above shower vent",
  "noteText": "Photo taken after landlord said the issue was fixed.",
  "capturedAt": "2026-03-27T18:10:00.000Z",
  "uploadedAt": "2026-03-27T18:15:00.000Z",
  "isDeleted": false
}
```

Name	Type	Description
_id	string	A globally unique identifier for the evidence record.

violationId	string	The id of the violation this evidence belongs to.
propertyId	string	The id of the related property.
uploadedByUserId	string	The id of the user who uploaded the evidence.
evidenceType	string	Type of evidence such as photo or note.
fileName	string	Name of the uploaded file.
fileUrl	string	Path or URL to the stored file.
mimeType	string	MIME type of the uploaded file, such as image/jpeg.
fileSize	number	Size of the uploaded file in bytes.
caption	string	Short description of the evidence.
noteText	string	User-entered note describing the evidence.
capturedAt	date	When the evidence was captured.
uploadedAt	date	When the evidence was uploaded.
isDeleted	boolean	Indicates if the evidence has been soft deleted.

5. Reviews

The Reviews collection stores structured ratings submitted by tenants for properties they have lived or live in. Each review captures three scored dimensions (responsiveness, value, and resolution) which are averaged into an overall score. Reviews are tied to both a property and a landlord allowing the application to show ratings from both perspectives. This collection directly supports the Rating System, the landlord trust score displayed on property profiles and the filtering and sorting available that will be on the violation table.

```
{
  "_id": "67e5b1d9a8c41234abcd5001",
  "propertyId": "67e5b1d9a8c41234abcd2001",
  "reviewerId": "67e5b1d9a8c41234abcd3001",
  "landlordId": "67e5b1d9a8c41234abcd4001",
  "responsiveness": 2,
  "value": 3,
  "resolution": 2,
  "overallScore": 2.3,
```

```
  "reviewText": "Slow to respond to calls.",
  "isDeleted": false,
  "createdAt": "2026-03-27T00:00:00.000Z",
  "updatedAt": "2026-03-27T00:00:00.000Z"
}
```

Name	Type	Description
_id	string	A globally unique identifier for the review
propertyId	string	The Id of the property being reviewed
reviewerId	string	The Id of the tenant user who submitted the review
landlordId	string	The Id of the landlord associated with the property at the time of review
responsiveness	number	Score from 1 5 rating how quickly the landlord responds to issues
value	number	Score from 1 -5 rating the overall value of the rental
resolution	number	Score from 1- 5 rating how effectively violations or issues were resolved
overallScore	number	Computed average of responsiveness, value, and resolution rounded to the 1st decimal
reviewText	string	Free-text field for the tenant to describe the review
isDeleted	boolean	Indicates if the review has been soft deleted
createdAt	date	Date the review was submitted
updatedAt	date	Date the review was last

		edited

6. Comments

This collection stores free-form messages posted to a property's public comment thread. Both tenants and landlords can participate, enabling a bath-and-forth communication that is visible to the general public (other users). Comments are linked to a property and record the author's role so the interface can visually distinguish tenant posts from landlord responses. This collection supports the social engagement feature described in our project proposal.

```
{
  "_id": "67e5b1d9a8c41234abcd6001",
  "propertyId": "67e5b1d9a8c41234abcd2001",
  "authorId": "67e5b1d9a8c41234abcd3001",
  "authorRole": "tenant",
  "commentText": "The mold in the bathroom has been getting worse",
  "isDeleted": false,
  "createdAt": "2026-03-27T00:00:00.000Z",
  "updatedAt": "2026-03-27T00:00:00.000Z"
}
```

Name	Type	Description
_id	string	A globally unique identifier for the comment
propertyId	string	The Id of the property this comment belongs to
authorId	string	The Id of the user who posted the comment
authorRole	string	Role of the author at the time of the posting: [tenant, landlord, admin]

commentText	string	The body of the comment
isDeleted	boolean	Indicates if the comment has been soft deleted
createdAt	date	Date the comment was posted
updatedAt	date	Date the review was last edited

7. Notifications

This collection will be centralized on the push-style alerts for notifications through email, text and certified letters.

```
{
  "_id": "notif-aa11bb22-cc33-dd44-ee55-ff6677889900",
  "userId": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "violationId": "viol-00112233-4455-6677-8899-aabbccddeeff",
  "channel": "sms",
  "notificationDetails": {
    "text": "Reminder: Mold violation at 111 Mold Street is due within 7
days."
  },
  "status": "queued",
  "createdAt": "2026-01-01-0000"
}
```

Name	**Type**	**Description**

_id	Object	A globally unique identifier to represent the notification
userId	string	Identifier for recipient users
channel	string	Delivery method sms, email or certified letter
notificationDetails	string	Contains details for delivery information ex. Mold violations due in 7 days
status	string	Status queued, sent, delivered, canceled, failed
createdAt	date	Creation time of the notification

8. Disputes

This collection is focused on the status management of a dispute and whether it has been remediated, resolved, or disputed.

```
{
  "_id": "dsp-aaaa-bbbb-cccc-ddddddddddd",
  "violationId": "viol-00112233-4455-6677-8899-aabbccddeeff",
  "createdBy": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "eventType": "status_change",
  "Status": "In-Progress",
  "Result": "Repair scheduled for 01-01-2026",
  "createdAt": "2026-01-01-0000"
}
```

Name	Type	Description

_id	Object	A globally unique identifier to represent the dispute
violationID	string	Identifier for the violation
createdBy	string	ID of user who created dispute
eventType	string	Notifier of change with the dispute
status	string	Status queued, sent, delivered, canceled, failed
Result	string	Current results/status of the dispute
createdAt	date	Creation time of the dispute

9. Attorneys

This collection will contain the list of referred attorneys that will highlight their personal and contact information.

```
{
  "_id": "refferal-legal-aid-001",
  "Name": "John Smith Legal",
  "email": "JohnSmith@NYC.legal",
  "phone": "+12125550000",
  "website": "https://johnsmithNYClegal.com",
```

```
  "createdAt": "2026-01-01-0000"
}
```

Name	Type	Description
_id	Object	A globally unique identifier to represent the referred attorney
Name	string	The name of the attorney
Email	string	Email contact for the attorney
phone	string	Phone number contact for the attorney
website	string	Business website for the attorney
createdAt	date	Creation time of the attorney profile

Group 24:

Daniel Park
Sarah Robinson
James Zangari
Chandler Tu